



FIRST SEMESTER, 2026–2027
Course Handout Part II

Date: 1st August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : MF F217
Course Title : Machine Drawing
Instructor-in-Charge : Prof. Amit R. Singh
Instructors : Mr. Aswanth V. Utham

1. Course Description:

Fundamentals of machine drawing and dimensioning practice using conventional software, orthographic drawing of machine parts, sectional view, assembly drawing, exploded view, pictorial view, BOM, materials assignment and graphic rendering.

2. Scope & Objective of the Course:

The objective of the course is to impart practice of machine drawing and solid modeling using a conventional 3D CAD software. The course will also cover Geometric Dimensioning and Tolerancing (GD&T).

3. Textbooks:

TB1. *Engineering Drawing & Design*, Cencil Jensen, Jay D. Helsel, and Dennis R. Short, 7th, Tata McGraw-Hill, 2012

References:

Ref1. *Visualization, Modeling and Graphics for Engineering Design*, Lieu D. K. and Sorby S., Cengage Learning, 2009

4. Course Plan:

Practical no.	Topic	Learning Outcomes
1- 3	Sketching	To construct fully constrained 2D geometrical sketches.
4 - 14	Solid modeling	- To learn constructive solid geometry operations like extrusion, revolution, sweeping, and lofting - To perform geometric transformations like boolean operations, mirror reflection, and creating patterns - To create holes, counterbores, counter-sinks, and threads - To create construction geometry
15 - 19	Assembly modeling	- To assemble components using joints - To perform interference, and motion studies
20 - 24	Geometric Dimensioning and Tolerancing	- To create accurate orthographic views - To create sectional views to show hidden features - To mark dimensions, and tolerances
25 - 26	Rendering	To create realistic renderings of 3D models

5. Evaluation Scheme:

Component	Wt (%)	Duration	Date & Time	Comments
Lab assignments	30	110 min	Lab hours	Open Book, best 4 of 6
Mid-Semester test	30	90 mins.	To be announced	Closed Book
Comprehensive Examination	40	180 mins.	To be announced	Closed Book

NC Criterion: Skipping all the lab assignments, the mid-semester exam or the final exam will result in a NC grade.

6. Evaluation Calendar:

Lab Session	Date	Topic
1	06-Aug-26	
2	07-Aug-26	
3	13-Aug-26	Sketching Test
4	14-Aug-26	
5	20-Aug-26	
6	21-Aug-26	Solid Modeling 1
7	27-Aug-26	
8	03-Sep-26	
9	10-Sep-26	
10	11-Sep-26	Solid Modeling 1
11	24-Sep-26	
12	25-Sep-26	
13	01-Oct-26	Midsemester Exam
14	15-Oct-26	
15	16-Oct-26	Assembly
16	22-Oct-26	
17	23-Oct-26	
18	29-Oct-26	Assembly and Motion 1
19	05-Nov-26	
20	06-Nov-26	
21	12-Nov-26	
22	13-Nov-26	
23	19-Nov-26	Geometric Dimensioning and Tolerancing
24	20-Nov-26	
25	26-Nov-26	
26	27-Nov-26	Comprehensive Exam

7. Office Consultation Hour:

Tuesday 4 pm to 4:50 pm. We can also set up a time by appointment via email.

8. Make-up Policy:

Make-up policy for the Mid-semester exam and the comprehensive exam will be as per AUGS guidelines. For lab assignments, no make-up will be allowed.

Prof. Amit R. Singh
Instructor-in-charge
MF F217 Machine Drawing